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Definition 11.1. *Meromorphic functions are "determined" by zeros and singularities.*
A point singulaity of f is a point $z_0 \in \mathbb{C}$ such that f is defined in a neighborhood of z_0 but not at z_0.

$$\mathcal{D}_\delta(z_0) \setminus \{z_0\}.$$

Example 11.2. *$f(z) = \frac{1}{z}$ has a singularity at $z_0 = 0$.*

Remark 11.3. *Zeros of a holomorphic f are isolated, unless $f \equiv 0$.*

Theorem 11.4. *(Local Description Near Zeros) Let $f : \underset{open}{\Omega} \to \mathbb{C}$ holomorphic and $f(z_0) = 0$, $z_0 \in \Omega, f \not\equiv 0$ then $\exists \mathcal{U} \ni z_0$, $g : \mathcal{U} \to \mathbb{C}$ holomorphic and vanishing ($\forall z \in \mathcal{U}, g(z) \neq 0$) $\exists! n > 0$ such that $f(z) = (z - z_0)^n g(z)$.*

Proof. We know

$$f(z) = \sum_{k=0}^{\infty} a_k (z - z_0)^k.$$

$f \not\equiv 0$ implies that $n =$ smallest k such that $a_k \neq 0$. We can write $f(z) = (z - z_0)^n g(z)$. □

Definition 11.5. *Let $n =$ the multiplicity/order of z_0. When $n = 1$, z_0 is called a simple zero. We say $f : \mathcal{D}_\delta(z_0) \setminus \{z_0\} \to \mathbb{C}$ has a pole at z_0 if $\frac{1}{f}$ extended by 0 at z_0 is holomorphic in $\mathcal{D}_\epsilon(z_0)$, for some $0 < \epsilon < \delta$.*

Theorem 11.6. *(Local Discontinuity near Poles) Let $f : \Omega \setminus \{z_0\} \to \mathbb{C}$ has a pole at z_0 in Ω then $\exists \mathcal{U} \ni z_0$, $\exists h : \mathcal{U} \xrightarrow{holo} \mathbb{C}$ nonvanishing then $\exists! n > 0$ such that $f(z) = (z - z_0)^{-n} h(z)$.*

Proof. Apply previous theorem to $\frac{1}{f}(z) = (z - z_0)^n g(z)$. □

Theorem 11.7. *(Laurent Series Expansion) If f has a pole of order n at z_0 then locally*

$$f(z) = \underbrace{\frac{a_{-n}}{(z - z_0)^n} + \cdots + \frac{\overbrace{a_{-1}}^{\text{residue of } f \text{ at } z_0}}{z - z_0}}_{\text{principal part}} + \underbrace{G(z)}_{\text{holo part of } f}.$$

where $G(z)$ is holomorphic and nonvanishing near z_0.
Moreover, $res_{z_0}(f) = \frac{1}{(n-1)!} \lim_{z \to z_0} \frac{d^{n-1}}{dz^{n-1}}((z - z_0)^n f(z))$

Proof. By theorem 11.6,

$$f(z) = (z - z_0)^{-n} h(z) = (z - z_0)^{-n} \sum_{k=0}^{\infty} b_k (z - z_0)^k = \frac{b_0}{(z - z_0)^n} + \cdots + \frac{b_{n-1}}{(z - z_0)} + \sum_{k \geq n}^{\infty} b_k (z - z_0)^{k-n}.$$

So we have

$$f(z)(z - z_0)^n = b_0 + b_1(z - z_0) + \cdots + b_{n-1}(z - z_0)^{n-1} + \underbrace{O((z - z_0)^n)}_{\to 0 \text{ as } \frac{d^{n-1}}{dz^{n-1}}}.$$

□

Theorem 11.8. *(The Residue Formula) Suppose $f : \Omega \setminus \underbrace{\{z_1, z_2, \cdots, z_n\}}_{poles} \to \mathbb{C}$ is holomorphic. Then for a disk $\overline{\mathcal{D}} \subset \Omega$ containing $z_1, ..., z_n$ we have*

$$\int_{\partial \mathcal{D}} f = 2\pi i \sum_{k=1}^{n} res_{z_k}(f).$$

Proof. Let $\gamma = \partial \mathcal{D}$ be the boundary of the disk $\mathcal{D}$, oriented counterclockwise. For each pole z_k, choose a small disk $\mathcal{D}_k$ centered at z_k with radius $\varepsilon_k > 0$ small enough so that:

- $\overline{\mathcal{D}_k} \subset \mathcal{D}$ for all k

- $\overline{\mathcal{D}_k} \cap \overline{\mathcal{D}_j} = \emptyset$ for $k \neq j$
- f is holomorphic on $\overline{\mathcal{D}} \setminus \bigcup_{k=1}^{n} \mathcal{D}_k$

Let $\gamma_k = \partial \mathcal{D}_k$ be the boundary of each small disk, oriented clockwise (negative orientation). Consider the multiply connected region $\mathcal{D} \setminus \bigcup_{k=1}^{n} \overline{\mathcal{D}_k}$.

By Cauchy's theorem for multiply connected regions, the integral of f around the boundary of this region (taking into account orientations) is zero. The boundary consists of γ (counterclockwise) and each γ_k (clockwise), so:

$$\int_{\gamma} f(z)dz + \sum_{k=1}^{n} \int_{\gamma_k} f(z)dz = 0.$$

Note that γ_k has clockwise orientation, so reversing it gives:

$$\int_{\gamma} f(z)dz = \sum_{k=1}^{n} \int_{-\gamma_k} f(z)dz.$$

where $-\gamma_k$ denotes γ_k with counterclockwise orientation.
Now, for each pole z_k, consider its Laurent expansion:

$$f(z) = \sum_{j=-\infty}^{\infty} a_j^{(k)}(z - z_k)^j = \frac{a_{-1}^{(k)}}{z - z_k} + \sum_{j \neq -1} a_j^{(k)}(z - z_k)^j.$$

where $a_{-1}^{(k)} = \operatorname{res}_{z_k}(f)$.
For $j \neq -1$, the function $(z - z_k)^j$ has an antiderivative on $\mathbb{C} \setminus \{z_k\}$ (or on all of $\mathbb{C}$ if $j \geq 0$), so by the fundamental theorem of calculus:

$$\int_{-\gamma_k} (z - z_k)^j dz = 0 \quad \text{for } j \neq -1.$$

For $j = -1$, we compute directly. Parameterize $-\gamma_k$ by $z(t) = z_k + \varepsilon_k e^{it}$ for $t \in [0, 2\pi]$:

$$\int_{-\gamma_k} \frac{1}{z - z_k} dz = \int_0^{2\pi} \frac{1}{\varepsilon_k e^{it}} \cdot i\varepsilon_k e^{it} dt = \int_0^{2\pi} i dt = 2\pi i.$$

Therefore,

$$\int_{-\gamma_k} f(z)dz = a_{-1}^{(k)} \cdot 2\pi i = 2\pi i \cdot \operatorname{res}_{z_k}(f).$$

Combining all terms:

$$\int_{\partial\mathcal{D}} f(z)dz = \sum_{k=1}^{n} \int_{-\gamma_k} f(z)dz = 2\pi i \sum_{k=1}^{n} \operatorname{res}_{z_k}(f).$$

□

12 Feb 11

Example 12.1.

$$\int_{-\infty}^{\infty} \frac{e^{ax}}{1+e^x} dx = \frac{\pi}{\sin(a\pi)}.$$

Proof. Let $f(z) = \frac{e^{az}}{1+e^z}$ then we have simple poles at $z : e^z = 1$ so $z \in \{..., -\pi i, \pi i, 3\pi i, ...\}$. We have the residual for

$$\int_{\gamma_1} f + \cdots = \int_{\gamma_R} f(z)\, dz = 2\pi i \operatorname{res}_{\pi i}(f).$$

We have

$$\int_{\gamma_1} f(z)dz \xrightarrow{R\to+\infty} \int_{-\infty}^{\infty} \frac{e^{ax}}{1+e^x} dx = I.$$

On γ_3, $z = x + 2\pi i$ with x from R to $-R$, so

$$\int_{\gamma_3} f(z)\, dz = -\int_{-R}^{R} \frac{e^{a(x+2\pi i)}}{1+e^x} dx = -e^{2\pi a i} \cdot I.$$

As $R \to \infty$, $\int_{\gamma_2} f \to 0$ and $\int_{\gamma_4} f \to 0$. Thus

$$(1 - e^{2\pi a i})I = 2\pi i \cdot \operatorname{res}_{\pi i}(f) = 2\pi i \cdot (-e^{a\pi i}) = -2\pi i e^{a\pi i},$$

$$\left|\int_{\gamma_2} f(z)dz\right| = \left|\int_0^{2\pi} \frac{e^{a(R+iy)}}{1+e^{R+iy}} i dy\right| \le \int_0^{2\pi} \frac{e^{aR}}{e^R+1} dy \xrightarrow{R\to\infty} 0$$

Additionally $\int_{\gamma_4} f \to 0$. To compute the residue at πi, we have

$$\lim_{z\to\pi i} f(z)(z - \pi i) = \lim_{z\to i\pi} \frac{e^{az}}{\frac{e^z - e^{\pi i}}{z - \pi i}} = \frac{e^{\pi i a}}{e^{\pi i}}.$$

Letting $R \to +\infty$ in the residual formula gives

$$I + 0 + 0 - e^{2\pi a i} I = -2\pi i e^{a\pi i}.$$

So

$$I = -\frac{2\pi i e^{\pi i a}}{1 - e^{2\pi a i}} = \frac{\pi}{\sin(a\pi)}.$$

□

Definition 12.2. *Let $f : \Omega \setminus \{z_0\} \to \mathbb{C}$ be holomorphic, if we can extend f analytically to z_0 we say that f has a removable singularity at z_0.*

Theorem 12.3. *(Riemann's Theorem) Let $f : \Omega \setminus \{z_0\} \to \mathbb{C}$ be holomorphic, if f is bounded near z_0 then z_0 is a removable singularity.*

Proof. Let $\overline{D} \subset \Omega$ be a disc centered at z_0. We want to use Cacuhy's formula

$$\frac{1}{2\pi i} \int_{\partial D} \frac{f(w)}{w - z} dw = g(z), \text{ for all} z \in D.$$

It suffices to show that $f = g$ on $D \setminus \{z_0\}$ because then g is the desired extension of f.
$F(w, z)$ is jointly continuous on $\partial D \times \overline{D}_{r-\epsilon}(z_0)$ and $\forall w \in \partial D$, $z \mapsto F(w, z)$ is holomorphic on D.
Fix $z \in D \setminus \{z_0\}$ then

$$\int_\gamma \frac{f(w)}{w-z} dw = 0 \implies \int_{\partial D} \frac{f(w)}{w-z} dw = \int_{C_\epsilon(z_0)} \frac{f(w)}{w-z} dw + \int_{C_\epsilon(z)} \frac{f(w)}{w-z} dw = 0 + 2\pi i f(z).$$

$\square$